

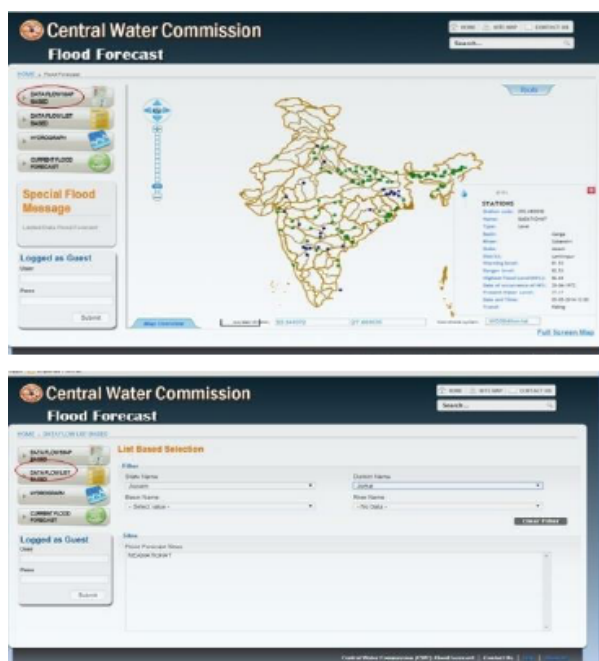
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Flood Forecasting Stations in India; Source:
Central Water Commission

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INTEGRATED FLOOD DISASTER MANAGEMENT AND SPATIAL INFORMATION: CASE STUDIES OF NETHERLANDS AND INDIA

[Conference Paper](#)[Full-text available](#)

Dec 2014

Sisi Zlatanova · Tarun Ghawana · Amarjeet Kaur · Jeroen M M Neuvel

Spatial Information is an integral part of flood management practices which include risk management & emergency response processes. Although risk & emergency management activities have their own characteristics, for example, related to the time scales, time pressure, activities &

actors involved, it is still possible to identify at least one common...

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Context in source publication

Context 1

... Measures Since 1954, 33,928 km of new embankments and 38,809 km of drainage channels have been constructed in the country. In addition, 2,450 town protection works have been completed and 4,721 villages have been raised above flood levels. Barring occasional breaches in embankments, these works have given reasonable protection to an area of about 18.22 million hectares. Non-structural Measures The MOWR and CWC had circulated in 1975, the draft bill for flood plain zoning regulations to the state governments for enactment and enforcement. The CWC has established a flood forecasting system comprising 175 stations on all major interstate rivers and is implementing the scheme for its modernisation and expansion. Some identified gaps by the guidelines are: □ Non-availability of close contour maps and digital elevation models of flood prone areas as impediments to make reliable forecasts. □ Flood forecasting and warning (FF and W) network of the CWC, though developed on scientific basis, does not cover all the flood prone rivers and rivulets. □ Very few river basins have been covered with automatic sensors for observations and telemetry system for communication of data. □ Flood plain zoning regulations have not been enacted and enforced by most of the states, □ There is also a lack of inflow forecasting systems for the reservoirs. The work of flood forecasting and warning in India is entrusted with the Central Water Commission (CWC). Flood Forecasting and flood warning in India was commenced in a small way in the year 1958 with the establishment of a unit in the Central Water Commission (CWC), New Delhi, for flood forecasting for the river Yamuna at Delhi. Presently, there are 878 Hydrological and Hydro-meteorological sites being operated by CWC across the country covering 20 river basins for gauge, discharge, sediment & water quality observations. The formulation of a forecast requires effective means of real time data communication network from the forecasting stations (Refer to Figure 1) and the

base stations (380 number approx. at present). Wireless Communication system installed in almost 550 stations is the backbone of the communication system required for flood forecasting activities. The activity of flood forecasting comprises of Level Forecasting and Inflow Forecasting. The level forecasts help the user agencies to decide mitigating measures (Flood Management). Figure 2 show its schematic diagram. It shows the horizontal and vertical functional relationships between the data collecting stations as well as various line departments on state and national level as flood forecasting centers & control rooms as well as users of the forecast. Press, Radio, TV, Police etc are also part of this schematic diagram. Figure 3 A shows the interactive data flow observations locations, which can be clicked upon to get the attribute data of a particular location. Figure 3 B shows the data extraction window. It provides options for query based selection of data using (spatial / thematic) state, district, basin or river name as filter. The main lesson derived from the exploration of flood risk and emergency management activities is that information has to flow across and between different actors involved in different flood management practices. A spatial information infrastructure that covers both risk and emergency management allows the required exchange of information. Therefore, the development of a spatial information infrastructure that integrates risk and emergency management deserves extra attention. These statements are discussed in more detail in a comparison between the characteristics of flood management and those of informational requirements. Figure 3 A & B: Data Flow Map and Data Flow List Based Selection Windows of CWC Flood Forecast ; Source: Central Water Commission The exploration of flood management activities in Netherlands has shown that flood risk and emergency management activities profoundly differ. Emergency management activities are often carried out under high time pressure and ask for tactical skills, whereas flood risk management activities are carried out under less time pressure and ask for strategic skills. Within both flood risk and emergency management, many stakeholders are involved on different scale / levels. As shown in our exploration, emergency preparedness and response is often the domain of emergency units such as medical services, the fire department and police services, while mitigation and recovery are often the domains of the planning, engineering and public works

departments. The exploration, however, has revealed a changing perspective on flood management, especially in the field of flood risk management. Within Dutch flood risk management, more attention is paid to flood consequences and possibilities for emergency management. In this light, flood risks are managed in a more integrated way in which both hazards, exposure, capacities and processes are taken into account. Land use planning was/is increasingly being considered an important instrument to reduce flood consequences. As a result, flood risk management is expected to become a more important issue in land use planning. In India, Disaster Management is not on an advanced stage and there is no clear-cut distinction of risk and emergency management concepts in practice. Flood Management in India using spatial information is at nascent stage although various actors are doing several activities for flood management. This include dedicated Flood Management Guidelines by National Disaster Management Authority (NDMA), State and District Level Authorities local level planning and actions, Satellite Images based observation, early warning and situation analysis by National Remote Sensing Centre, Research conducted by various disaster management related organizations. Indian Flood Management guidelines stress on spatial information for different phases of activities as well as to plug the identified gaps. The Flood Forecast System of the Central Water Commission has a Graphical User Interface (GUI) which provides spatial as well as statistical information on click of a station in Data Flow Map Window. Data Flow List Window allows user to filter the query on different spatial scale of administrative boundaries (State, District) or natural boundaries / Features (Basin, River). Further, this allows to directly select Forecasted Sites which again provides spatial as well as statistical information. The informational requirements of two stakeholders involved in two different domains of flood and emergency management have been discussed in more detail, as in the case of Netherlands: the land use planning department and the fire department. Even though the activities of the land use planners and the fire department differed, it was interesting to observe that informational requirement considerably overlapped. The informational requirements with respect to the static and simulated information

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Nov 2014

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Spatial Information is an integral part of flood management practices which include risk management & emergency response processes. Although risk & emergency management activities have their own characteristics, for example, related to the time scales, time pressure, activities & actors involved, it is still possible to identify at least one common...

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... Flood is one of the serious natural disasters all over the world [1] [2] [3][4]. Its impact has been intensified by increasing human settlements in flood-prone areas [5]. ...

... Although the investigations demonstrated that using the soft computing algorithms in detecting shelters and safe areas have been well-organized, some significant deficiencies in these studies include: (1) all fundamental risk factors (spatial and environmental) for temporary sheltering in flood have not been considered, (2) to apply the temporary shelters as evacuation stations, the optimal distribution of them is vital, while it has been ignored in related studies. Due to the heterogeneous infrastructure

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 Najmeh Neysani Samani · Mahdi Sheybani ·  Sisi Zlatanova

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... Applications of GIS ranges from risk analysis of industrial and chemical accidents (Renjith and Madhu, 2010), to analysis and decision-making support in extreme weather and flooding scenarios. Zlatanova et al. (2014) explores the current state of GIS and spatial information within flood disaster management, and outlines the current and future roles of spatial information within the domain. Examples of GIS in flood disaster analysis show that GIS and hydrological tools can be used to model disasters (Al-Abadi et al., 2016), and to convey risk and consequences of extreme weather disasters to the general population (Artigas et al., 2016). ...

... The last decade has seen dramatic improvements in GIS and ICT technology, creating the technical backbone in the form of SDIs and improving the spatial capabilities of DSS Fabbri, 2009, Stoimenov et al., 2007). These developments paves the way for integration of the components

required to increase the utilisation of GIS beyond analysis of static information in emergency management organisations (Zlatanova et al., 2014). ...

... Spatial information and tools should to a larger degree be shared between organisations participating in planning and response efforts to ensure a shared situational understanding. Zlatanova et al. (2014) demonstrates the relation between access to information in the planning and response phases. An examination of the Dutch and Indian flood disaster management documented the increasing overlap in informational requirements, and suggested a focus on multipurpose in spatial information and GIS. ...

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... Developed countries, where anthropic settlements determined extensive urbanization processes, have to deal with an existing building stock located in hazard prone areas which is not always «hazards proofed» and is ageing: such built assets require the implementation of mitigation and protection measures against natural threads that are continuously evolving due to the current global climate change. Several references provide lists of parameters to be considered in the evaluation of vulnerability (Federal Emergency Management Agency, 2007) (Van Westen et al. 2008) (Zlatanova

et al., 2014) and fine grain analysis on buildings vulnerability are not new in the scientific panorama (Sucuoğlu et al., 2007) (Ernst et al., 2010) (Stephenson et al., 2014). ...

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... Developed countries, where anthropic settlements determined extensive urbanization processes, have to deal with an existing building stock located in hazard prone areas which is not always «hazards proofed» and is ageing: such built assets require the implementation of mitigation and protection measures against

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... India, one of the tropical countries, has the greatest problem of being severely affected by floods (Sharma et al. 1996). The flood risk management and emergency response process need an integrated platform for data/information gathering, risk assessment, appraisal of options and making, implementing, and reviewing decisions to reduce, control, accept, or redistribute risks of flooding (Hall et al. 2003).The spatial information is now an integral part of flood management (Zlatanova et al. 2014). Satellite remote sensing technology provides expeditious and accurate tempo- ral information about flood extent helpful in long-term planning. ...

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... Third, since the course of a crisis is often unpredictable, data requirements and the constellation of involved stakeholders might suddenly change. This might lead to an overload of the central crisis management system as it has to integrate constantly changing datasets (Zlatanova et al., 2014). Fourth, current systems access and display all the data available not differentiating between stakeholders' roles, which require different kinds of information (Zlatanova et al., 2014). ...

... This might lead to an overload of the central crisis management system as it has to integrate constantly changing datasets (Zlatanova et al., 2014). Fourth, current systems access and display all the data available not differentiating between stakeholders' roles, which require different kinds of information (Zlatanova et al., 2014). This may create an information overload since the human cognitive processing capacity is limited (Endres et al., 2015). ...

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